

**DA BALI RESORT DIGITAL TRANSFORMATION:
WEB-BASED BOOKING AND RESERVATION
MANAGEMENT SYSTEM**

**SUBMITTED TO THE FACULTY OF THE
DEPARTMENT OF INFORMATION TECHNOLOGY**

BY

EMIE R. GANDIONGCO

JIGGY O. MACARAYO

PRAISE GAE TANQUI-ON

**IN PARTIAL FULFILLMENT OF THE
REQUIREMENTS FOR
THE DEGREE OF

BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY**

JUNE 2026

CERTIFICATE OF ORIGINALITY

I hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material to which to a substantial extent has been accepted for award of any other degree or diploma of a university or other institute of higher learning, except where due acknowledgement is made in the text.

I also declare that the intellectual content of this capstone project is the product of our work, even though I may have received assistance from others on style, presentation and language expression.

EMIE R. GANDIONGCO

JIGGY O. MACARAYO

PRAISE GAE TANQUI-ON

Signature of advisee over printed name

Date

NEVEL TAURINE C. AUXILIO, MIT

Signature of advisor over printed name

Date

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ACRONYMS AND ABBREVIATIONS

- **Admin:** Administrator
- **IEC:** International Electrotechnical Commission
- **IJISRT:** International Journal of Innovative Science and Research Technology
- **IJRASET:** International Journal for Research in Applied Science & Engineering Technology
- **ISO:** International Organization for Standardization
- **MIT:** Master of Information Technology
- **MSEUF:** Manuel S. Enverga University Foundation
- **PIDS:** Philippine Institute for Development Studies
- **RSIS:** Research and Scientific Innovation Society
- **USA:** United States of America
- **ERD:** Entity Relationship Diagram
- **FDD:** Functional Decomposition Diagram
- **IPO:** Input-Process-Output
- **PMS:** Property Management System
- **POS:** Point of Sale
- **RBAC:** Role-Based Access Control
- **SDLC:** Software Development Life Cycle
- **SUS:** System Usability Scale
- **UAT:** User Acceptance Testing
- **UI:** User Interface
- **API:** Application Programming Interface
- **BPI:** Bank of the Philippine Islands
- **CSS3:** Cascading Style Sheets 3
- **ENUM:** Enumerated Type
- **FAQ:** Frequently Asked Questions
- **FK:** Foreign Key

- **HTML5:** HyperText Markup Language 5
- **JSON:** JavaScript Object Notation
- **JWT:** JSON Web Token
- **LTS:** Long-Term Support
- **PDF:** Portable Document Format
- **PK:** Primary Key
- **RDBMS:** Relational Database Management System
- **RFID:** Radio-Frequency Identification
- **SMS:** Short Message Service
- **ARM:** Advanced RISC Machines
- **CPU:** Central Processing Unit
- **DDR4:** Double Data Rate 4 Synchronous Dynamic Random-Access Memory
- **GB:** Gigabyte
- **GHz:** Gigahertz
- **HD:** High Definition
- **INT:** Integer
- **Mbps:** Megabits per second
- **RAM:** Random Access Memory
- **SSD:** Solid State Drive
- **VARCHAR:** Variable Character

CHAPTER I

INTRODUCTION

1.1 Background of the Study

Resorts in foreign areas like the USA, such as those in tropical Hawaii, sunny Florida, and scenic California coastlines, have surged in popularity over the past decade, providing luxurious getaways with amenities like beachfront villas, spa treatments, infinity pools, and adventure excursions. These destinations attract global travelers seeking relaxation and rejuvenation amid demanding urban lives (Dalal, 2025).

Cebu resorts often struggle with manual booking processes like phone calls and social media messages, leading to delays, errors, and even scams where tourists arrive without valid reservations (SunStar Cebu, 2026; Cebu Insights, 2025). These fragmented methods create operational chaos in hotspots like Bantayan Island and Moalboal, where over 200 scam cases were reported since late 2025 due to fake online agents demanding e-wallet payments (Philstar, 2026). A centralized booking website for Cebu could automate reservations, verify listings, and provide real-time availability, boosting trust and competing with bigger chains (Cebu Insights, 2025).

Across the Philippines, the digital divide persists as many resorts rely on outdated "brick-and-mortar" tactics amid slow tourism recovery, risking double bookings and lost revenue against regional competitors (PIDS, 2025; Metrobank Wealth Insights, 2024). Small operators in remote areas face fragmented guest records and no dynamic pricing, worsened by scams mimicking legitimate channels (RSIS International, 2025). A national

centralized reservation portal would unify inventory, enable instant confirmations, and connect eco- lodges to global travelers for inclusive growth (RSIS International, 2025).

Specifically in the coastal beauty of Balingasag, spots like Da Bali Resort in Barangay Rosario serve as vital landmarks for local relaxation. However, managing a resort in such a high-demand area requires more than just a scenic view; it requires precision. Relying on manual coordination for room availability and amenity scheduling in a busy municipality means the staff spends more time on paperwork and less time ensuring the guests in Barangay Rosario are having the best experience possible.

The solution lies in a centralized web-based platform. By migrating from a manual "wait-and-see" system to a real-time reservation website, Da Bali Resort can offer 24/7 accessibility to its guests. A centralized system doesn't just "save a spot"; it organizes the entire resort ecosystem—from tracking payments to managing peak-hour traffic—allowing the management to see their operations through a clear, data-driven lens rather than a cluttered desk of receipts.

Ultimately, the goal is to bridge the gap between traditional Balingasag hospitality and modern technological efficiency. By implementing a dedicated website, Da Bali Resort sets a new standard for the local industry in Misamis Oriental. It transforms the administrative burden into a seamless digital handshake, ensuring that from the first click in an app to the final check-out at the resort, the guest's peace of mind is never interrupted by manual errors.

1.2 Conceptual Framework

This study follows the Input-Process-Output (IPO) paradigm, which divide the system development into three interconnected logical phases as illustrated in Figure 1.1 to ensure the structured development of the Da Bali Resort business platform. This framework ensures that guest information is accurately captured, processed through robust logical safeguards to prevent errors, and transformed into high-value outputs for both the guest and the resort management.

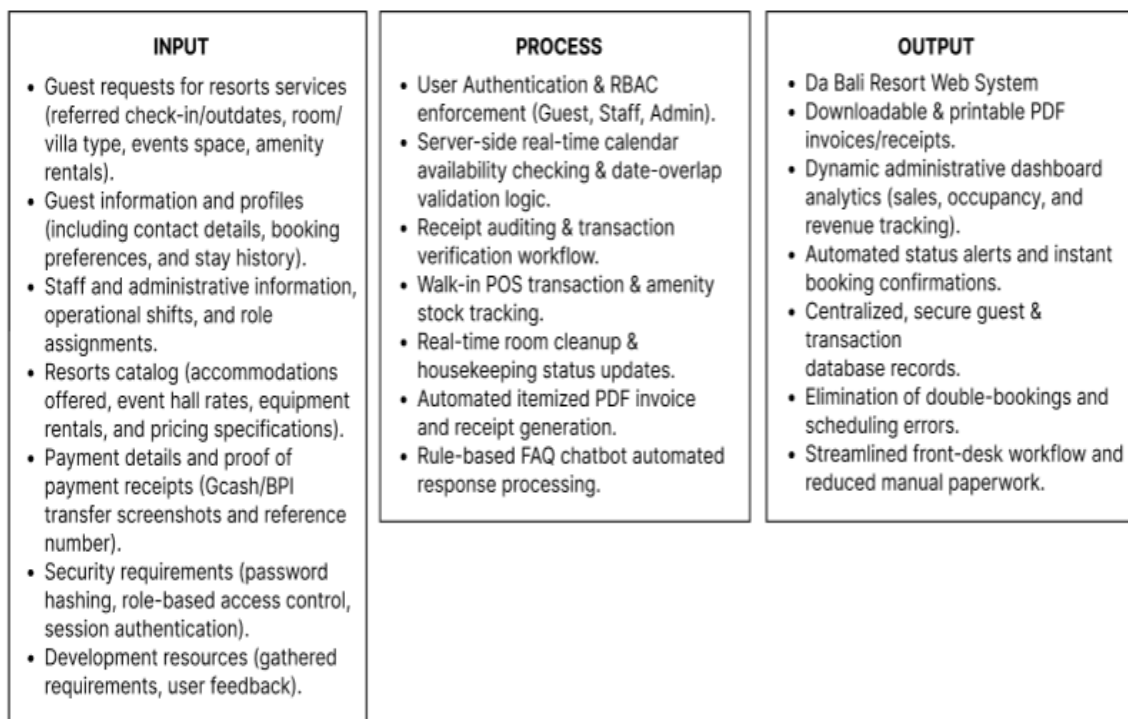


Figure 1.1 IPO Conceptual Framework

Figure 1.1 illustrates the conceptual framework of the study using the Input-Process-Output (IPO) model, mapping out the systematic transformation of operational data into a fully integrated digital management system for Da Bali Resort.

Input

The Input encompasses all the essential data, credentials, specifications, and development resources required to operate and construct the system. It includes guest service requests such as stay dates, room/villa types, event halls, and amenity rentals, alongside guest account profiles and contact details. On the administrative side, it captures staff credentials, shift schedules, role assignments, and the complete resort catalog containing accommodation rates and amenity rental pricing. Furthermore, it incorporates proof-of-payment details—specifically GCash or bank transfer screenshots and reference numbers—as well as system security specifications like Bcrypt password hashing and session tokens. Finally, it integrates software development resources, including user requirements and SDLC feedback.

Process

The Process represents the core logical, administrative, and automated functions executed by the application itself. It begins with User Authentication and Role-Based Access Control (RBAC) enforcement across Guest, Staff, and Admin roles to ensure data security. To maintain operational accuracy, the system runs server-side real-time calendar availability checks and date-overlap validation logic to prevent scheduling conflicts. Front-desk operations are handled through a receipt auditing and transaction verification workflow, paired with a walk-in POS transaction and amenity stock tracking mechanism. The system also manages real-time room cleanup and housekeeping status updates, automates itemized PDF invoice and receipt generation, and processes user queries through a rule-based automated FAQ chatbot response engine.

Output

The output of the study is the successful development of the Da Bali Resort. The system provides essential features and generated outputs, such as downloadable and printable PDF invoices and receipts, dynamic administrative dashboard analytics for tracking sales, occupancy, and revenue, automated status alerts with instant booking confirmations, and centralized, secure guest and transaction database records. Through these features, the system achieves key operational outcomes, including the total elimination of double-bookings and scheduling errors, alongside a streamlined front-desk workflow with reduced manual paperwork. The study also produces system evaluation results based on the responses of selected respondents to assess software quality and usability. Based on these findings, recommendations for further system enhancement and future improvements are provided to support the continuous development and implementation of the proposed application. CertDe

1.3 Statement of the Problem

1.3.1 General Problem

Despite the growing demand for resort experiences in the Philippines, local establishments continue to grapple with outdated manual processes that undermine operational efficiency and guest satisfaction. Specifically, Da Bali Resort struggles with manual logging, fragmented communication, and delayed booking confirmations. To resolve these operational challenges, this study addresses the following specific issues:

1.3.2 Specific Problem

- **Inquiries and Communication Lag:** Slow response times and fragmented communication channels (such as phone calls and social media messages) result in lost bookings from impatient travelers.
- **Scheduling Conflicts and Record Losses:** A high incidence of double-bookings and lost customer records occurs due to reliance on physical logbooks and paper-based tracking across staff shifts.
- **Financial and Payment Inefficiencies:** Revenue losses stem from unverified payments, no-shows, overbookings, and slow manual receipt auditing during high-demand periods.
- **Administrative Workload Overhead:** Increased staff workload from manual paperwork diverts time and labor away from on-site guest services and operational precision.
- **Limited Digital Accessibility:** A lack of 24/7 digital accessibility limits scalability and competitiveness for modern travelers seeking instant online booking convenience.

1.4 Objectives of the Study

1.4.1 General Objective

The primary objective of this study is to design, develop, and evaluate a centralized web-based booking and reservation management platform – Da Bali Resort Digital Transformation: Web-Based Booking and Reservation Management System – to modernize resort operations, eliminate manual inefficiencies,

and enhance guest satisfaction through real-time digital solutions.

1.4.2 Specific Objective

- **Automated Communication & Support:** To integrate an automated rule-based FAQ support bot and instant booking status alerts to streamline inquiries and reduce guest abandonment.
- **Overlap Validation & Inventory Management:** To engineer a centralized database featuring server-side date-overlap validation logic and real-time availability tracking to eliminate double-bookings and ensure accurate record-keeping.
- **Financial Tracking & Invoicing Engine:** To integrate digital payment tracking modules (including GCash/bank receipt uploads) and an automated itemized invoicing system to minimize payment delays and financial losses.
- **Centralized Operational Dashboard:** To construct a unified Admin/Staff web dashboard equipped with real-time inventory controls and point-of-sale (POS) tracking to eliminate physical paperwork and reduce staff workload.
- **Responsive Web Platform Development:** To build a fully responsive, mobile-optimized web interface that provides 24/7 availability viewing, positioning the resort competitively in the digital travel market.

1.5 Significance of the Study

The implementation of the Da Bali Resort Digital Transformation: Web-Based Booking and Reservation Management System provides substantial operational, technical, and financial benefits to various key stakeholders. By replacing manual paper logbooks with a centralized, automated web platform, this study directly addresses existing workflow bottlenecks and creates value for the following groups:

For the Resort Guests. Provides a seamless, self-service digital experience that allows 24/7 real-time availability checking, instant submission of proof-of-payment receipts, automated pricing transparency, and instant answers to common inquiries through an automated FAQ assistant. This eliminates long inquiry wait times and reduces booking friction.

For the Resort Manager. Delivers comprehensive operational oversight through centralized admin dashboards, automated itemized billing, real-time inventory tracking, and reliable financial reporting. It enables data-driven business decisions, minimizes revenue loss from unverified payments or double-bookings, and improves overall business competitiveness.

For the Front-Desk Staff. Significantly reduces administrative workload by eliminating tedious physical logbook entries, shift-transition ledger errors, and manual payment auditing. The centralized point-of-sale (POS) and automated date-overlap prevention logic allow staff to process walk-in and online reservations efficiently while focusing on high-quality guest service.

For Future Researchers and System Developers. Serves as a practical architectural reference and empirical benchmark for

modernizing small-to-medium hospitality enterprises (SMEs) in provincial regions. It offers reusable system designs, normalized database schemas, and validation logic algorithms that can be adapted for similar web-based reservation studies.

1.6 Scope and Limitations

This study focuses on the design, development, and usability evaluation of the Da Bali Resort Digital Transformation: Web-Based Booking and Reservation Management System tailored specifically for Da Bali Resort located in Balingasag, Misamis Oriental. The primary scope encompasses core functional components including secure user management with role-based access control (Admin, Staff, and Guest) utilizing Bcrypt password encryption and session handling, as well as a real-time availability and booking engine featuring dynamic calendar views for accommodations (villas and cottages), function halls, and amenity rentals. To maintain scheduling integrity, the application incorporates a server-side date-overlap validation engine engineered to eliminate double-bookings and operational collisions across all resort offerings.

Furthermore, the system integrates a payment and receipt verification module that enables guests to upload proof-of-payment receipts, such as GCash or bank transfer screenshots with reference numbers, for front-desk administrative auditing. Operational management is further supported by itemized PDF invoicing, a Point-of-Sale (POS) desk interface tailored for walk-in guests, and an automated rule-based FAQ assistant designed to answer frequent guest inquiries regarding rates, amenities, and policies. The technical quality and user satisfaction of the overall system are empirically evaluated using the System Usability Scale (SUS) and the ISO/IEC 25010 Quality Model across a sample of 10 respondents, comprising 1

resort manager, 2 front-desk staff, and 7 end-user guests.

Despite these capabilities, several operational constraints and boundaries define the specific limitations of the project. The platform does not feature real-time automated payment gateway APIs, such as direct PayMaya or Stripe API webhooks, for instant credit or debit card processing; instead, financial verification relies entirely on guest-uploaded payment receipts manually verified by front-desk personnel. Additionally, because the application is built as a web-based management system, it requires a stable internet connection for real-time database synchronization and calendar updates, offering no support for offline data synchronization.

Finally, the software is designed strictly as a responsive web application accessible through standard mobile and desktop web browsers, meaning dedicated native mobile application builds (such as iOS .ipa or Android .apk files) remain outside the current scope of development. The system also operates independently of physical resort infrastructure, as it does not interface directly with hardware components such as electronic RFID door locks, automated keycard dispensers, or physical turnstiles.

1.7 Definition of Terms

To facilitate a clearer understanding of the terms used in this study, the following variables and technical terms are defined conceptually and operationally:

Administrative Dashboard

Refers to a graphical user interface that aggregates and displays key performance indicators, operational data, and system metrics in a centralized location for streamlined monitoring and decision-

making. In this study, it represents the back-end managerial interface of the Da Bali Resort system that provides authorized staff and administrators with real-time analytics on sales, room occupancy rates, revenue metrics, and booking status overviews.

Bcrypt Algorithm

Refers to a key-derivation function based on the Blowfish cipher used to securely hash and store passwords. In this study, it serves as the primary cryptographic measure applied to guest and administrative account passwords before saving them into the system database, ensuring protection against unauthorized access and data breaches.

GCash Receipt Verification

Refers to the administrative process of cross-referencing and validating electronic proof-of-payment receipts uploaded by users against official transaction reference numbers and amounts. In this study, it denotes the verification module within the Da Bali Resort system that allows resort administrators to review uploaded GCash payment screenshots and transaction reference numbers submitted by guests to authenticate reservations before final booking confirmation.

Iterative Model

Refers to a software development process that breaks the overall system development into small, repeatable cycles, allowing the platform to be built, tested, and refined progressively. In this study, it represents the specific Software Development Life Cycle (SDLC) framework utilized to incrementally design, prototype, and refine the Da Bali Resort web-based system based on continuous testing and stakeholder feedback.

ISO 25010 Software Quality Standard

Refers to an internationally recognized software evaluation framework that defines product quality characteristics such as functional suitability, performance efficiency, usability, reliability, security, and maintainability. In this study, it serves as the evaluation standard used to assess the overall technical quality, performance, and functionality of the Da Bali Resort web-based booking system based on feedback gathered from respondents.

Online Reservation System

Refers to an internet-based software solution that allows users to view available services, make bookings, and complete transactions digitally without physical presence. In this study, it denotes the web application designed specifically for Da Bali Resort that enables guests to check real-time availability, select accommodations or amenities, upload payment proofs, and receive automated booking confirmations.

Point of Sale (POS) System

Refers to a combination of software and hardware used to record retail transactions, compute order totals, and process payments at the point of service. In this study, it refers to the front-desk administrative module within the platform that allows resort staff to handle walk-in guest transactions, track physical amenity rentals, and manage onsite payments efficiently.

Real-Time Calendar Synchronization

Refers to an automated data-updating mechanism that dynamically reflects changes, bookings, and availability across a system immediately without requiring manual page refreshes or delays. In this study, it denotes the server-side logic in the Da Bali Resort

application that continuously updates accommodation and amenity availability, validating date overlays in real time to prevent scheduling conflicts and eliminate double-bookings across online and walk-in reservation channels.

Role-Based Access Control (RBAC)

Refers to a security mechanism that restricts system access based on defined roles assigned to authorized users. In this study, it defines the permissions and dashboard interfaces designated for Guests, Staff, and Administrators within the Da Bali Resort system, ensuring users only access features relevant to their specific roles.

System Usability Scale (SUS)

Refers to a standardized, reliable ten-item Likert scale tool used to evaluate the overall usability and user satisfaction of a software product. In this study, it is the evaluation instrument administered to end-users and resort stakeholders to measure the ease of use, learnability, and overall user acceptance of the developed system.

CHAPTER II

REVIEW OF RELATED LITERATURE

This chapter presents the review of related literature and studies that provide the theoretical and empirical foundation of the present study. The reviewed literature discusses the major variables relevant to the development of the proposed Da Bali Resort System. It systematically examines the operational vulnerabilities inherent in manual paper-based booking practices, details modern software architectures used in building scalable web reservation systems, and analyzes the measurable business impact delivered by digital platforms in hospitality management. By synthesizing these perspectives into structured narrative discussions, this review provides a solid academic foundation and justifies the technical design of the proposed platform.

2.1 Related Literature

2.1.1 Foreign Literature

2.1.1.1 Manual Booking Issues

Relying on traditional paper ledgers and manual booking practices exposes hospitality businesses to frequent human errors, lost revenue, and operational inefficiencies. A study conducted by Hotelogix researchers (2021) highlighted that manual record-keeping at small resorts frequently results in double-bookings, revenue losses of up to 15%, and severe guest dissatisfaction during peak seasonal rushes. Building on this, an investigation by SiteMinder (2022) revealed that reliance on physical logbooks and phone calls increases reservation no-show rates by up to 25% due to the lack of automated confirmation alerts.

The operational strain caused by human error was further detailed by GraceSoft (2023), whose findings demonstrated that administrative staff waste approximately 30% of their daily working hours on repetitive manual ledger updates instead of attending to guest needs. In line with these findings, BookB (2025) reported that unmonitored scheduling overlaps erode guest trust and create daily administrative friction in hospitality operations.

Finally, Hayes and Miller (2022) established that non-automated tracking methods inhibit real-time room availability visibility, hindering efficient inventory distribution and compounding financial loss during periods of high demand.

Collectively, these foreign studies demonstrate that manual reservation practices consistently impair business performance, burden staff, and degrade guest satisfaction across small-scale lodging properties. Addressing these deep-seated operational bottlenecks highlights the critical necessity of transitioning from vulnerable paper logbooks to automated digital reservation management systems.

2.1.2 Centralized Web Technologies

Centralized web architecture and modern software stacks provide the technical framework needed to synchronize reservation data and automate daily resort workflows. According to Nguyen (2025), replacing legacy logbooks with a full-stack web application built using Vue.js for the responsive frontend, Node.js for the backend server, and MySQL for database management successfully transforms tourism operations by offering real-time updates,

automated confirmation notifications, and intuitive administrative dashboards.

Enhancing backend communication capabilities, IJISRT Authors (2023) demonstrated how utilizing the Laravel framework alongside integrated short message service (SMS) gateways facilitates reliable real-time guest notifications. In mobile-centered architectures, Chen and Wang (2024) analyzed modern digital platforms, emphasizing that intuitive user interfaces combined with social media feature integration significantly streamline booking interactions and enhance customer stay experiences.

Furthermore, IJRASET Authors (2025) proved that combining automated web portals with conversational chatbot assistants reduces routine front-desk administrative inquiries. Supporting these structural developments, Garcia and Lopez (2022) affirmed that decoupled API architectures allow concurrent database queries, preventing scheduling collisions and supporting continuous system scaling.

Overall, these technical studies underscore that adopting robust, decoupled full-stack frameworks and modern web APIs provides the essential infrastructure required for dynamic room validation, instant guest notifications, and real-time database synchronization. These proven architectural patterns directly inform the software design choices made for the proposed platform.

2.1.1.3 Benefits of Web Systems

Adopting automated web platforms drives business growth by expanding market reach, improving payment

convenience, and boosting overall operational performance. A study by Agyei et al. (2025) investigating 183 star-rated hotel properties in Accra Metropolis revealed that direct booking websites and online travel networks provide rapid payment processing (\$M=3.785\$), uninterrupted 24/7 accessibility (\$M=3.634\$), and optimized booking volume (\$M=3.639\$), solidifying web platforms as vital business revenue drivers.

Expanding on this, Garcia and Lopez (2022) revealed that digital platforms foster sustainable tourism management by streamlining administrative overhead beyond basic reservation convenience. From a financial perspective, Nazli (2020) showed that direct web reservation portals reduce reliance on high-commission third-party booking agents, thereby safeguarding profit margins and strengthening direct guest relationships.

Moreover, Softomate Solutions (2024) established that automated transaction confirmation tools dramatically lower customer drop-off rates through instant notifications. Lastly, Ciaramella (2022) concluded that real-time digital inventory tracking enables resort managers to maintain high occupancy rates while completely eliminating manual entry mistakes.

In summary, the empirical evidence confirms that implementing direct web booking platforms generates substantial economic and operational advantages, ranging from maximized revenue generation to enhanced customer retention. These global findings validate the strategic importance of digitalizing property management functions.

2.1.2 Local Literature

2.1.2.1 Local Manual Systems Challenges

Local tourism establishments across the Philippines face severe administrative difficulties due to disorganized paper records, missed social media inquiries, and overbooking risks. A report published by the Philippine Institute for Development Studies (PIDS, 2024) indicated that manual processes in local Philippine resorts lead to lost reservation requests originating from social media channels and cause frequent overbooking errors in regional tourist hubs like Balingasag.

This issue is further compounded in major destination areas, where SiteMinder (2022) noted that manual reliance on paper ledgers strains front-desk personnel in popular tourist areas like Cebu, where disorganized records elevate administrative errors and increase susceptibility to booking scams. Focusing on resort-level operations, Balucot et al. (2025) observed that Nature Spring Resort faced significant operational delays due to manual reservation tracking, inefficient cottage assignment, and unorganized record management.

In a similar study, D' Rock Resort Developers (2024) documented that peak-season walk-in queues and uncoordinated phone inquiries routinely overwhelmed front-desk staff, resulting in extended waiting times and missed business opportunities. Ultimately, Ramento et al. (2024) confirmed that fragmented manual logbooks used across regional resorts induce time-consuming errors that hinder long-term operational growth.

Together, these local investigations reveal that Philippine resorts suffer from distinct operational friction caused by reliance on unintegrated, paper-bound administrative practices. Resolving these localized challenges emphasizes the immediate requirement for tailored web-based solutions suited to local resort environments.

2.1.2.2 Impact on Local Tourism

Transitioning to localized digital reservation platforms directly strengthens the economic resilience, service quality, and competitiveness of regional Philippine tourism. The MSEUF Research Team (2023) demonstrated that equipping regional lodging providers with web portals featuring real-time analytics transforms traditional operations into highly competitive, data-driven enterprises.

Highlighting the macro-economic benefits, the PIDS (2024) report emphasized that implementing standardized digital booking systems across Philippine resorts reduces administrative overhead and prevents revenue loss caused by unrecorded booking inquiries on social media platforms.

In terms of customer satisfaction, Balucot et al. (2025) established that providing guests with direct online amenity selection and instant booking validation significantly improves user satisfaction scores and operational control for resort owners.

Expanding market reach, D' Rock Resort Developers (2024) proved that web-based platforms allow local resorts to reach customers beyond local walk-in visitors,

successfully attracting out-of-town tourists and increasing overall profitability. Finally, GraceSoft (2023) reinforced that reallocating resort personnel from manual administrative tasks to direct guest support dramatically improves hospitality service delivery and overall operational excellence.

Ultimately, the literature highlights that modernizing local reservation workflows yields positive economic ripple effects across the regional hospitality sector. By eliminating operational inefficiencies and expanding digital accessibility, localized booking platforms serve as vital catalysts for sustainable business growth in Philippine tourism.

Table 2.1 Existing Application

Application	Description	Key Strengths	Limitations
Agoda	A popular booking website widely used across Asia for hotel rooms.	• Very popular in Asia • Sends quick booking confirmation messages	• Charges high fees for every booking • Cannot handle hourly rentals for cottages or resort amenities • Hard for small resort staff to set up and manage
Cloudbeds	PMS + Channel Manager with dynamic pricing and guest messaging for mid-size properties	• Keeps bookings updated across multiple websites • Shows detailed sales reports	• Very expensive to pay for every month • Too complicated for regular resort staff to learn • Stops working properly if internet connection is lost

Booking.com	A famous global website used to book rooms, flights, and rental cars.	• Reaches travelers worldwide • Accepts credit card payments automatically • Includes customer review features	• Takes high commission fees (15%–25% per booking) • Does not support local e-wallets like GCash receipt uploads • Cannot manage walk-in guests at the front desk
Airbnb	A platform where owners list their houses or rooms for short stays.	• Very easy to use on mobile phones • Let's guests chat directly with the host • Handles payments safely	• Designed for house rentals, not resort day-use or cottages • Charges extra service fees to both owners and guests • Lacks a front-desk system for staff to manage resort items
Hotelogix	An online system designed to help small and medium hotels run front-desk operations.	• Simple dashboard for front-desk staff • Tracks room cleaning status • Handles different currencies	• Charges high monthly fees • Cannot check GCash receipt screenshots uploaded by guests • Has too many unnecessary features for simple cottage rentals
Little Hotelier	An all-in-one booking system made specifically for small inns and bed-and-breakfasts.	• Has a simple phone app for owners • Easy to set up and view calendar bookings • Has a direct booking page for websites	• Monthly payments add extra costs for the resort • Rules are too strict for local day-use or hourly rental policies • Has no tool to manually verify bank or e-wallet payment receipts

Table 2.1 presents a comparative evaluation of existing commercial booking and property management platforms commonly used in the hospitality industry, such as Booking.com, Agoda, Airbnb, Cloudbeds, Hotelogix, and Little Hotelier. It outlines each platform’s core functions, key operational strengths, and specific limitations when applied to local resort settings. This analysis highlights common gaps in commercial software—such as high commission costs, lack of support for local payment receipt uploads (e.g., GCash), and complex interfaces—thereby establishing the practical need for a custom, cost-effective web system designed specifically for Da Bali Resort.

2.2 Table 2.2 Comparison of Related Studies

Related Studies	Features						
	Online Booking	Real-Time Calendar	Receipt Verification	POS (Walk-In)	FAQ Chatbot	Role-Based Access	Reports & Analytics
Nguyen (2025)	✓	✓				✓	
Balucot et al. (2025)	✓	✓				✓	✓
D' Rock Resort Developers (2024)	✓			✓			✓
MSEUF Research Team (2023)	✓					✓	✓
PIDS Report (2024)							✓

Da Bali Resort System (Proposed)	✓	✓	✓	✓	✓	✓	✓
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The Table 2.2 provides a feature-by-feature matrix comparing previous local and foreign literature with the proposed Da Bali Resort System. The table evaluates key technical capabilities across the studies, including online booking, real-time calendar synchronization, payment receipt verification, walk-in Point-of-Sale (POS) management, automated FAQ chatbots, role-based access control, and reporting analytics. As illustrated in the matrix, while existing studies address individual components of reservation management, the proposed system uniquely integrates all seven core features into a unified platform tailored to the operational requirements of Da Bali Resort

CHAPTER III

METHODOLOGY

The methodology described herein presents the systematic approach required to design, develop, and evaluate the Da Bali Resort. This section provides a operational roadmap of the technical, architectural, and analytical processes involved, ensuring that the software platform is functional, secure, and specifically tailored to the operational demands of resort management in Barangay Rosario, Balingasag, Misamis Oriental. By mapping system requirements directly to architectural deployment and empirical evaluation, this methodology ensures high data integrity, eliminates double-booking errors, and provides a structured framework for validating system performance.

3.1 Research Design

This study adopts a descriptive developmental research design to engineer, implement, and evaluate the Da Bali Resort. The descriptive facet provides a structured approach for investigating current operational workflows, mapping manual bottlenecks, and documenting user requirements from both resort administrative personnel and prospective guests. Through direct observational analysis and stakeholder interviews at the resort in Barangay Rosario, Balingasag, the descriptive phase establishes the baseline parameters needed to formulate the platform's functional specifications—such as real-time date-overlap prevention, gated balance settlement rules, and walk-in POS capabilities.

To translate these identified needs into a working software product, the developmental component executes the systematic engineering, database normalization, and frontend-backend architecture of the system. Development is governed by an Iterative (Rapid Prototyping) Software Development Life Cycle (SDLC) model, enabling the project team to rapidly build functional software modules, expose them to end-users for hands-on evaluation, and continuously refine the codebase. This iterative feedback loop ensures that technical edge cases—including concurrent booking requests, receipt upload validation, and chat routing—are thoroughly tested and resolved prior to final deployment.

Following system implementation, the platform's overall technical performance and end-user acceptability are quantitatively evaluated using the ISO/IEC 25010 Software Quality Model alongside the System Usability Scale (SUS). The ISO/IEC framework establishes standardized metrics to measure functional suitability, operational usability, performance efficiency, security, and reliability across various client devices. By pairing a descriptive assessment of operational needs with an iterative prototyping methodology and standard quality benchmarks, this research design guarantees a robust, secure, and user-centric digital transformation for Da Bali Resort.

3.2 System Design

3.2.1 Iterative Software Development Life Cycle (SDLC) Model



Figure 3.1 Iterative Model for Da Bali Resort

Figure 3.1 depicts the Iterative Model utilized in the development and continuous refinement of the Da Bali Resort Digital Transformation: Web-Based Booking and Reservation Management System. This cyclic software engineering approach allows the system to undergo progressive enhancement through repeating loops, ensuring that functional requirements, database stability, and user feedback are addressed effectively before final system handover.

1. **Planning:** Establishes the foundational technical roadmap, resource allocation, and core architectural decisions needed to digitize Da Bali Resort's booking workflows using Vue.js, Node.js, and MySQL. It outlines operational boundaries and development schedules for each iterative increment.
2. **Requirements:** Gathers and specifies the functional needs and

business rules of resort administrators, staff, and end-user guests. Key specifications identified include real-time date-overlap prevention logic, proof-of-payment auditing, gated balance settlements, and front-desk walk-in POS needs.

3. **Analysis & Design:** Translates the gathered functional specifications into detailed technical blueprints and system artifacts. During this phase, system architecture layouts, normalized database schemas (ERD), data dictionaries, system flowcharts, and responsive user interface (UI) wireframes are designed.
4. **Implementation:** Transforms the design blueprints into functional, executable software components through active programming. Frontend user portals, backend API service endpoints, server-side calendar algorithms, and MySQL database tables are coded and integrated.
5. **Testing:** Evaluates the functional modules within a local environment to verify date-overlap validation routines, balance calculation logic, and session security. Rigorous testing identifies bugs, logic errors, or database conflicts to ensure system stability prior to stakeholder review.
6. **Evaluation:** Collects hands-on feedback from resort management, administrative staff, and target end-users through ISO/IEC 25010 quality metrics and System Usability Scale (SUS) assessments. Insights and user feedback gathered from this phase loop directly back into subsequent Planning and Requirements cycles for continuous system refinemen

3.3 System Architecture

The technical architecture of the Da Bali Resort Web System follows a decoupled, three-tier model designed to separate presentation logic, backend application processing, and data persistence. Figure 3.2 illustrates the overall system architecture, demonstrating the flow of data between client-side interactions, application controllers, and normalized database storage.

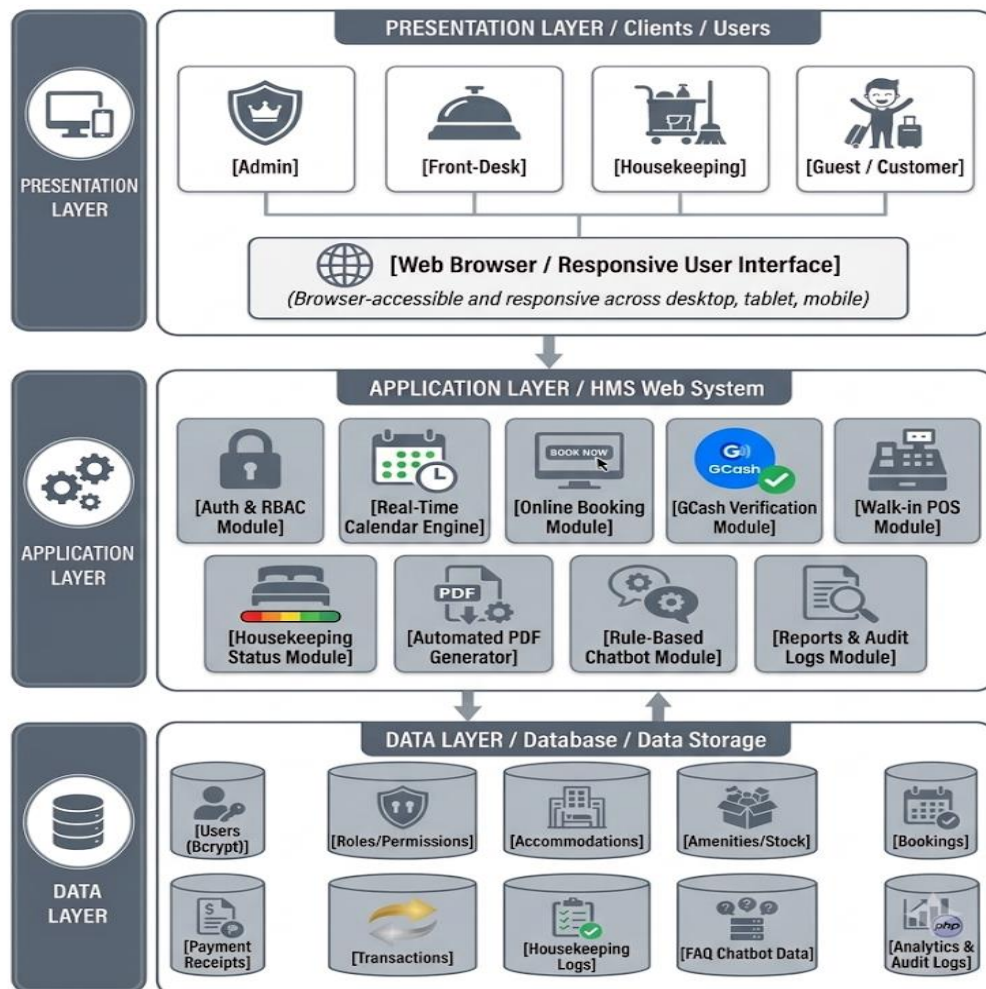


Figure 3.2 System Architecture

Presentation Layer (Clients / Users) serves as the front-end user interface where system actors interact with the application. The system supports four distinct user roles:

- **Admin:** Accesses high-level managerial features, reports, and system configuration controls.
- **Front-Desk:** Operates day-to-day front-desk workflows, walk-in transactions, and receipt checks.
- **Housekeeping:** Monitors and updates room cleanup statuses in real time.
- **Guest / Customer:** Browses accommodations, submits reservation requests, uploads proof-of-payment receipts, and interacts with the automated chatbot.

Application Layer (HMS Web System) serves as the central processing unit of the platform, executing application logic, workflow automation, and security enforcement. It consists of nine core system modules:

- **Auth & RBAC Module:** Enforces login authentication and Role-Based Access Control to restrict access based on user permissions.
- **Real-Time Calendar Engine:** Performs server-side scheduling checks and date-overlap validation logic to eliminate double-bookings.
- **Online Booking Module:** Processes guest reservation requests, room selections, and schedule bookings.
- **Payment Audit & Invoicing Engine:** Controls the gated workflow that holds room confirmation until staff verify receipt uploads, triggering PDFKit to generate automated itemized guest invoices upon final settlement.
- **GCash Verification Module:** Manages the auditing and

verification of uploaded GCash transaction screenshot receipts and reference numbers.

- **Walk-in POS Module:** Handles front-desk transactions for walk-in guests and tracks amenity rentals onsite.
- **Housekeeping Status Module:** Provides real-time room cleanup status updates and room maintenance tracking.
- **Automated PDF Generator:** Creates downloadable and printable itemized PDF invoices and transaction receipts dynamically.
- **Rule-Based Chatbot Module:** Automatically processes guest inquiries and delivers pre-configured FAQ responses.
- **Reports & Audit Logs Module:** Aggregates analytics on sales, room occupancy, and revenue while logging administrative actions.

Data Layer (Database / Data Storage) serves as the system's centralized database repository, handling data persistence and structured storage across ten dedicated data entities:

- **Users (Bcrypt):** Stores user credentials encrypted using the Bcrypt hashing algorithm.
- **Roles / Permissions:** Defines privileges assigned to Guests, Staff, Housekeeping, and Admins.
- **Accommodations:** Contains details, pricing, and specs for rooms, villas, and event spaces.
- **Amenities / Stock:** Tracks physical rental items and current inventory stock levels.
- **Bookings:** Records active, pending, confirmed, and historical reservation entries.
- **Payment Receipts:** Archives uploaded proof-of-payment screenshots and GCash reference records.

- **Transactions:** Logs financial payment details, balances, and invoice numbers.
- **Housekeeping Logs:** Stores history and real-time statuses of room cleaning tasks.
- **FAQ Chatbot Data:** Houses automated query templates, key phrases, and canned responses.
- **Analytics & Audit Logs:** Retains operational data, system activity histories, and managerial performance logs.

3.4 Context Diagram

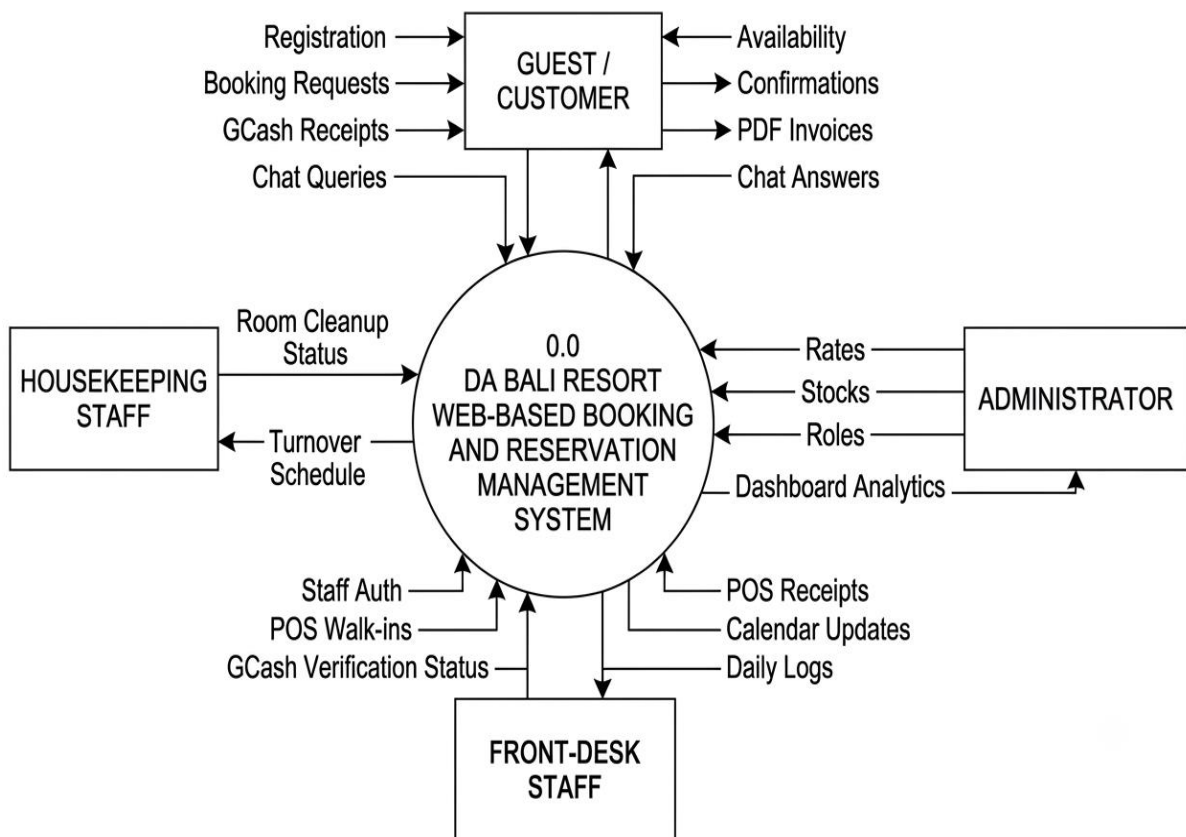


Figure 3.3 Context Diagram

The Context Diagram (Level 0 Data Flow Diagram) defines the global boundary of the Da Bali Resort Web System, illustrating the high-level inputs,

outputs, and data exchanges between external entities and the centralized system. Figure 3.3 illustrates the overall Context Diagram, mapping the interaction flow among the three primary external entities: Guest / Customer, Resort Staff / Front-Desk, and Resort Admin / Management.

Entity 1: Guest / Customer

Inputs to System (Arrows pointing TO System):

- Registration & Login Credentials
- Booking Requests & Accommodation / Amenity Selections
- Uploaded GCash Payment Receipts & Reference Numbers
- Chatbot Inquiries & FAQs

Outputs to Guest (Arrows pointing TO Guest):

- Real-Time Accommodation Availability & Rates
- Booking Status Alerts & Instant Confirmations
- Downloadable / Printable PDF Invoices & Receipts
- Automated Chatbot Responses

Entity 2: Front-Desk Staff

Inputs to System (Arrows pointing TO System):

- Staff Authentication Credentials
- Walk-in Guest Registration & POS Orders
- Onsite Payment Details & Amenity Rentals
- GCash Payment Receipt Verification Status (Approve / Reject)

Outputs to Front-Desk Staff (Arrows pointing TO Staff):

- Real-Time Reservation Calendar & Availability Updates
- Printable POS Walk-in Receipts & Invoices
- Daily Front-Desk Transaction Logs & Occupancy Overview

Entity 3: Housekeeping Staff

Inputs to System (Arrows pointing TO System):

- Housekeeping Authentication Credentials
- Room Cleaning & Maintenance Status Updates (e.g., Clean, Dirty, Under Maintenance)

Outputs to Housekeeping Staff (Arrows pointing TO Housekeeping):

- Daily Room Turnover List & Cleanup Schedule
- Checkout Alerts & Real-Time Room Inspection Requests

Entity 4: Administrator

Inputs to System (Arrows pointing TO System):

- Admin Credentials & Security Management
- System Configuration Settings (Rates, Room Types, Stocks)
- Account Management & Role Assignment

Outputs to Administrator (Arrows pointing TO Admin):

- Real-Time Dashboard Analytics (Sales, Occupancy, Revenue)
- System-wide Audit Logs & Activity History
- Generated Financial & Operational Reports

3.5 Functional Decomposition Diagram (FDD)

The Functional Decomposition Diagram illustrates the structural hierarchy of the Da Bali Resort Web System by breaking down the platform into its primary operational domains and supporting sub-modules. Figure 3.4 illustrates the overall Functional Decomposition Diagram, mapping the relationships among the platform's core functional areas.

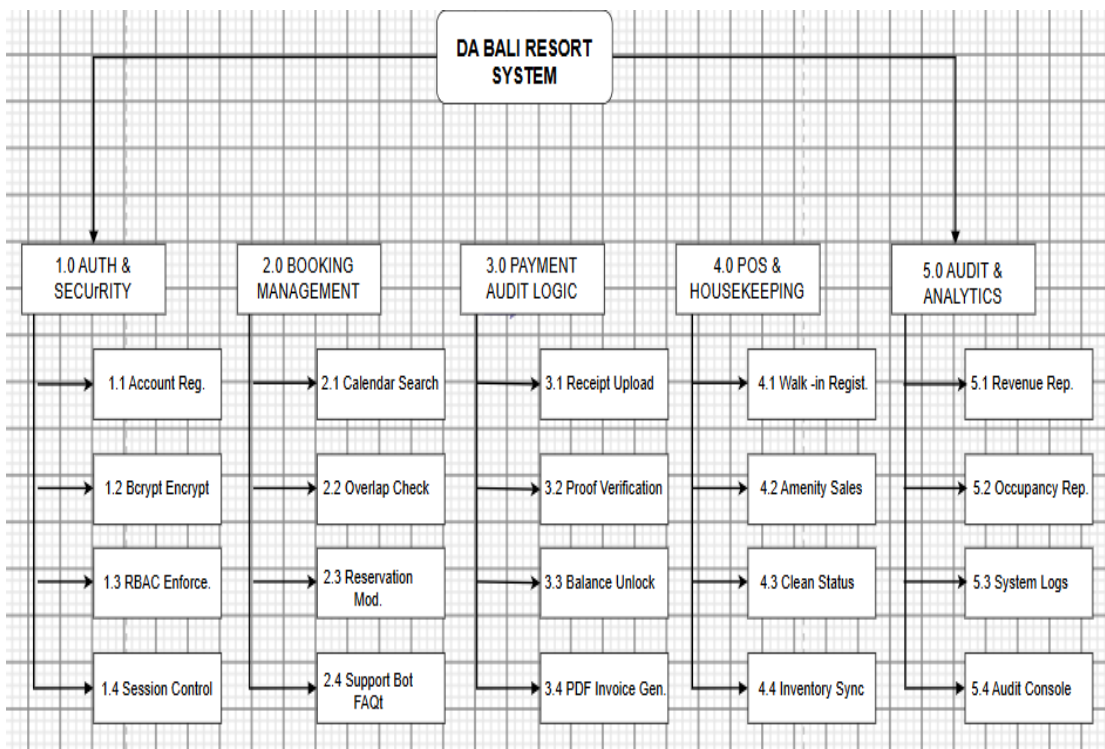


Figure 3.4 Da Bali Resort Functional Decomposition Diagram

At the core of client interactions are the Authentication and Security and Booking Management modules. The Authentication module handles registration, login validation, Bcrypt password hashing, session management, and Role-Based Access Control (RBAC) across Guest, Staff, and Admin accounts. Working alongside security, the Booking Management module allows guests to check interactive availability calendars, process stay reservations, run server-side date-overlap validation to prevent double-bookings, and interact with an automated FAQ support bot for quick inquiries.

The operational and backend workflows are driven by the Payment Audit Logic, Front-Desk POS & Housekeeping Tracker, and Operational Audit & Analytics modules. The Payment Audit module manages receipt image uploads, staff verification interfaces, gated balance releases, and automated PDF invoice generation. Meanwhile, the Front-Desk POS and Housekeeping module supports direct walk-in guest registrations, amenity rental sales, inventory updates, and real-time room cleanliness tracking. Finally, the Analytics module compiles administrative audit logs, occupancy statistics, and itemized financial reports for resort management.

3.6 Use Case Diagram

The Use Case Diagram models the system's operational boundaries and functional capabilities by illustrating interactions between external actors and primary system use cases. Figure 3.5 illustrates the overall Use Case Diagram, depicting how the Guest, Front-Desk Staff, and Resort Administrator actors interact with core features such as booking management, POS operations, payment audits, and analytics reporting.

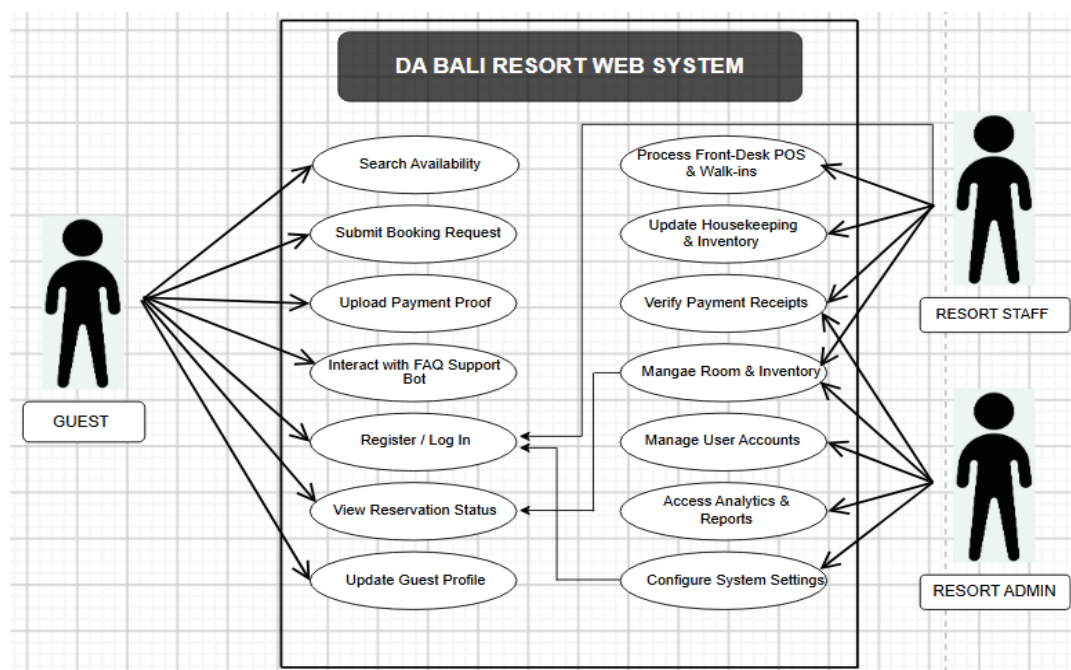


Figure 3.5 Use Case Diagram

The system interaction is driven by three primary actors: Guest, Resort Staff, and Resort Administrator. Guests can search accommodation availability, submit reservation requests, upload transaction receipts for payment verification, and interact with the automated FAQ support bot for quick inquiries. To maintain operational security, submitting booking requests automatically includes user authentication and triggers backend date-overlap validation to prevent double-bookings.

On the operational side, Front-Desk Staff handle walk-in guest registrations, process Point-of-Sale (POS) transactions for amenity rentals, and continuously update housekeeping statuses (e.g., Clean, Occupied, Maintenance) across resort facilities. Both Front-Desk Staff and Resort Administrators share access to the payment audit module to review uploaded guest receipts and trigger automated PDF invoice generation upon balance clearance. Additionally, Resort Administrators maintain full control over system configurations, account privileges, and dynamic financial/occupancy analytics reporting.

3.7 Entity Relationship Diagram

The Entity-Relationship Diagram models the logical database structure of the Da Bali Resort Web System, detailing data entities, attributes, primary/foreign key relationships, and cardinality constraints. Figure 3.6 illustrates the overall Entity-Relationship Diagram, mapping the database architecture across five normalized core entities.

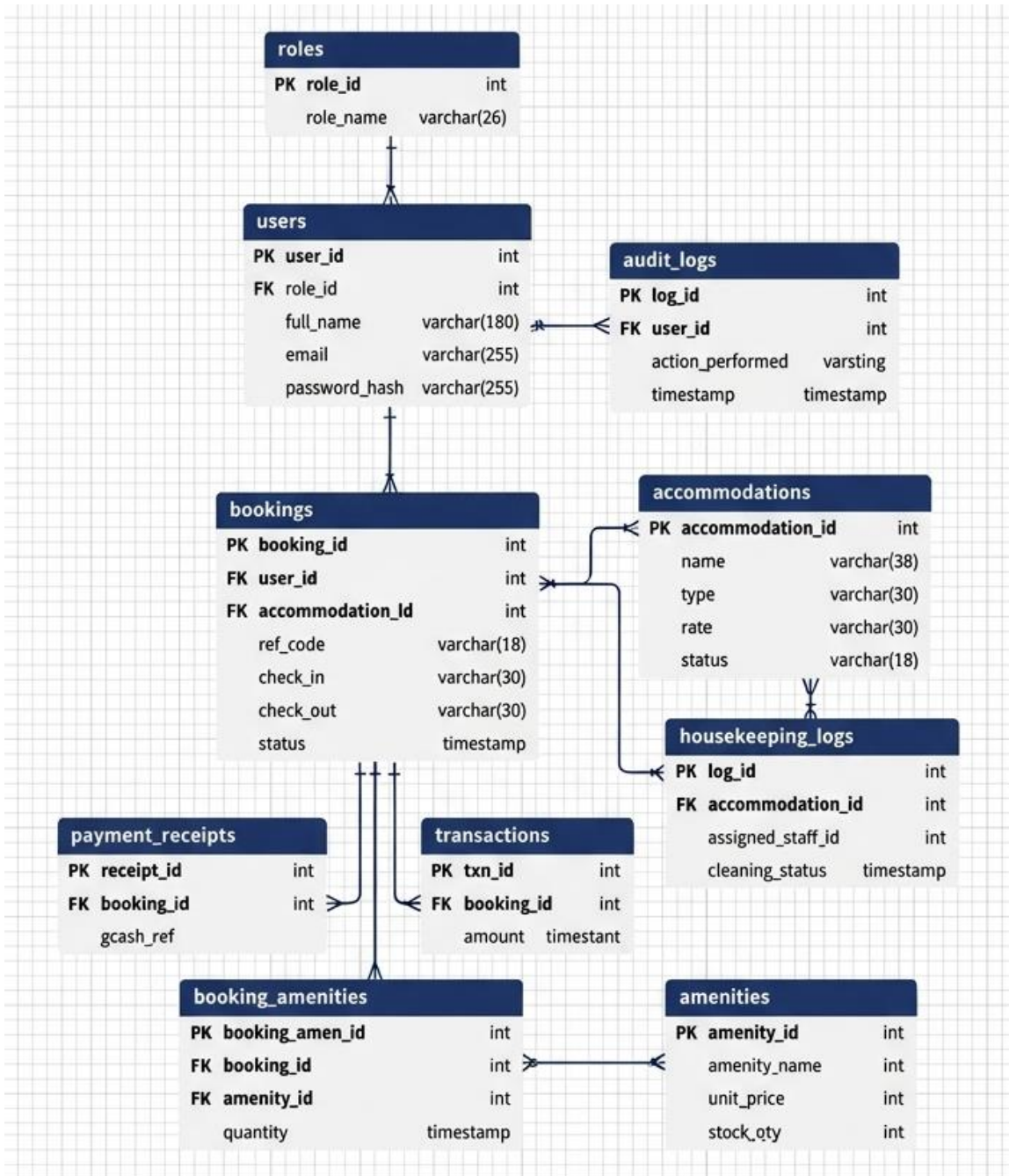


Figure 3.6 Entity Relationship Diagram

User and Access Management (USERS, ROLES): The ROLES entity enforces Role-Based Access Control (RBAC) across system actors (Admin, Front-Desk, Housekeeping, Guest) in a 1-to-Many relationship with the USERS entity. Sensitive user passwords stored in USERS are hashed using the Bcrypt algorithm.

Reservation Core (ACCOMMODATIONS, BOOKINGS): The BOOKINGS entity links USERS and ACCOMMODATIONS. Server-side calendar logic evaluates check-in and check-out dates within BOOKINGS to prevent date overlaps and double-bookings.

Amenity and POS Tracking (AMENITIES, BOOKING_AMENITIES): A Many-to-Many relationship between BOOKINGS and AMENITIES is resolved through BOOKING_AMENITIES, allowing walk-in and online guests to rent multiple amenities while automatically deducting from real-time stock levels.

Payment and Receipt Verification (PAYMENT_RECEIPTS, TRANSACTIONS): Proof-of-payment screenshots submitted by guests are logged in PAYMENT_RECEIPTS, where front-desk staff verify GCash reference numbers. Confirmed payments generate corresponding financial entries in TRANSACTIONS, which feed into printable PDF invoice generation and administrative revenue analytics.

Operations and Maintenance (HOUSEKEEPING_LOGS, AUDIT_LOGS): The HOUSEKEEPING_LOGS entity tracks room cleanup and turnover activities assigned to housekeeping staff. All critical system actions performed by users are persistently tracked in AUDIT_LOGS for security and administrative monitoring.

3.8 Flowcharts

3.8.1 System Flowchart

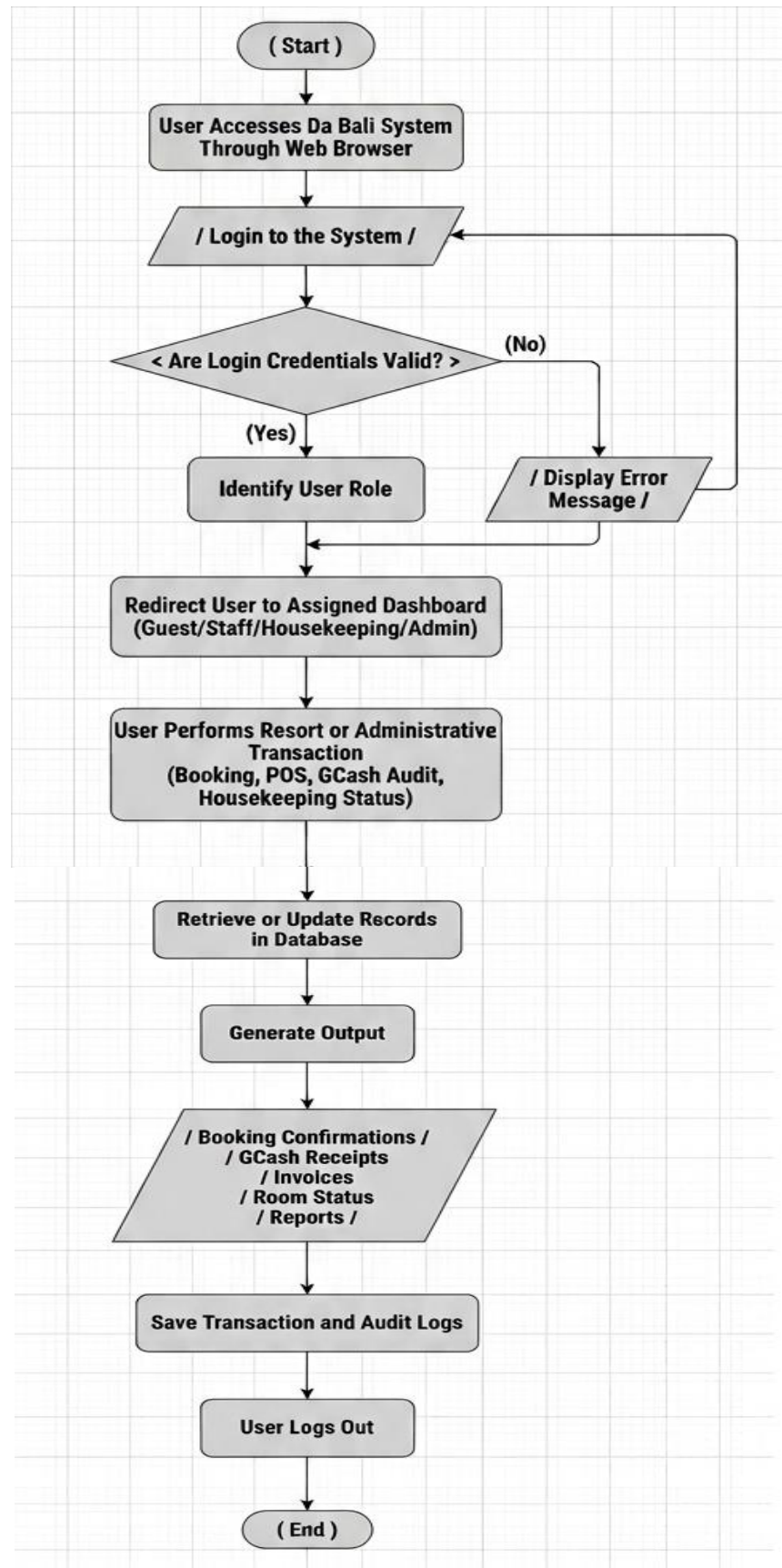


Figure 3.8 System Flowchart of Da Bali Resort

Figure 3.8 illustrates the sequential operational flow and data processing execution within the Da Bali Resort System. The flowchart details the end-to-end system logic from initial user access to final session termination across four main phases:

User Access and Authentication Phase: The system process initiates when a user accesses the platform via a web browser on a desktop or mobile device. Upon entering account credentials, the system evaluates their validity through a decision branch. If the credentials are invalid, the system displays an error message and redirects the user back to the login interface. If valid, the system identifies the user's specific role (Guest, Front-Desk Staff, Housekeeping, or Resort Administrator) and grants access by redirecting the user to their designated dashboard interface.

Transaction Execution Phase: Within their authorized dashboard, the user executes operational or administrative transactions. Depending on the logged-in role, these transactions include submitting online room reservations, uploading GCash proof-of-payment receipts, managing walk-in POS sales, updating housekeeping cleanup statuses, or generating financial and occupancy reports.

Data Retrieval and Output Generation Phase: Connected via off-page connector A, the system queries and updates the corresponding tables within the central database. Following processing, the system generates dynamic outputs, including booking confirmation statuses, GCash payment verification alerts, downloadable PDF transaction receipts, updated real-time calendar availability states, and administrative summary analytics.

Audit Logging and Session Termination Phase: Upon transaction completion, the application automatically writes and saves system logs, financial records, and user activity history into the persistent

database for security and auditing purposes. The process concludes when the user explicitly logs out, safely terminating the system session.

3.8.2 Program Flowchart

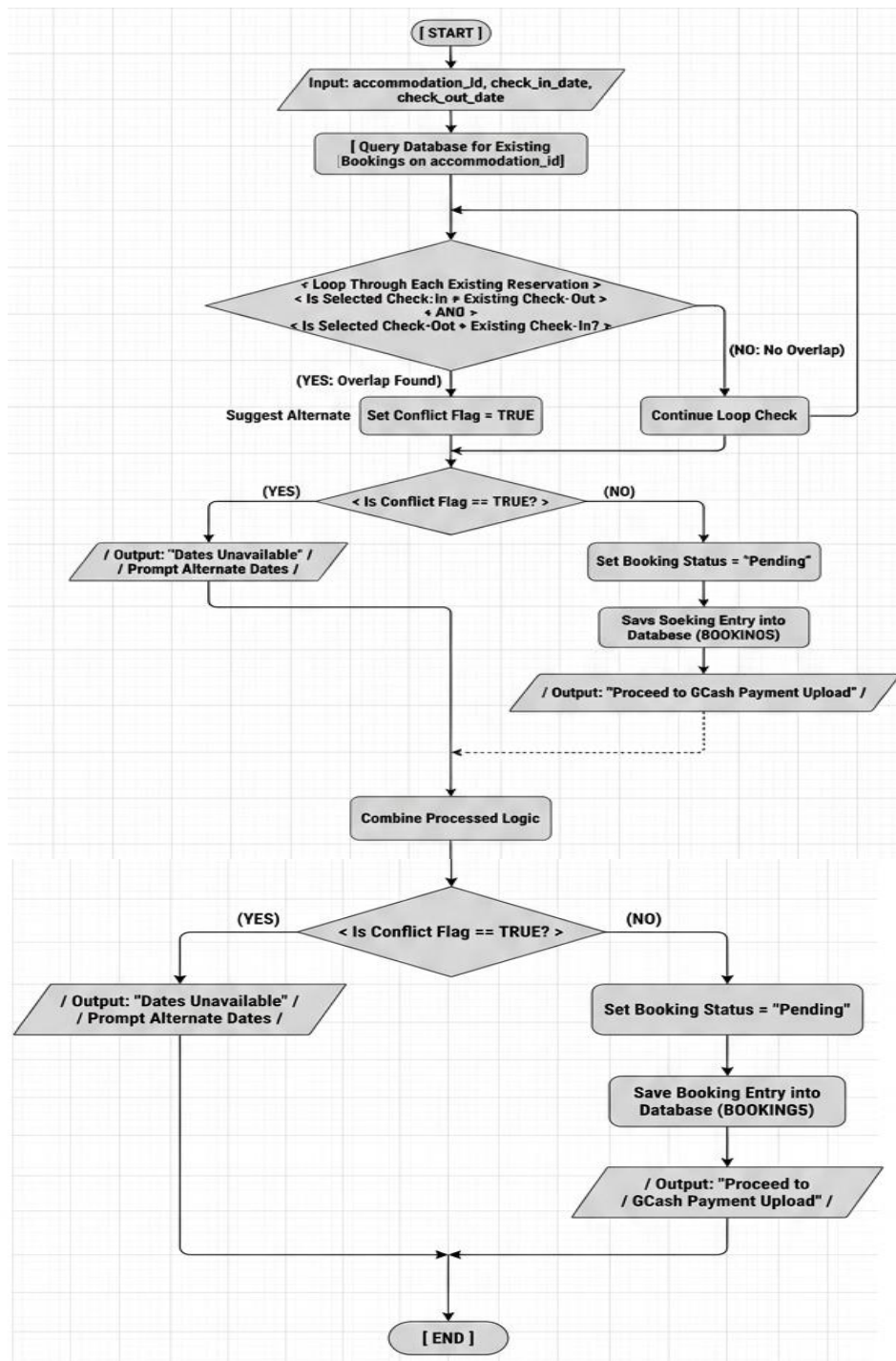


Figure 3.9 Program Flowchart of Da Bali Resort

Figure 3.9 illustrates the detailed low-level programmatic logic and algorithmic execution flow of the Real-Time Calendar Synchronization and Date-Overlap Validation Engine. The process begins when the user inputs the target accommodation ID, requested check-in date, and requested check-out date. The system executes a database query to retrieve all active reservation entries associated with the specified unit, preparing the dataset for iterative evaluation.

The algorithm then executes a loop evaluation through each existing reservation record to check for scheduling collisions. Using the overlap logic condition, the program checks if the selected check-in date is prior to an existing check-out date while the selected check-out date extends past the existing check-in date. If an overlap is detected (YES), the system sets a conflict flag to TRUE and prompts alternative options. Conversely, if no collision is detected (NO), the program continues the loop check until all existing bookings are evaluated.

Finally, the program evaluates the state of the conflict flag to finalize the transaction. If the conflict flag is TRUE, the system outputs a "Dates Unavailable" notification and prompts the user to select alternate dates. If the conflict flag is FALSE, the system sets the booking status to "Pending," saves the new reservation entry directly into the database's BOOKINGS table, and prompts the user to proceed to the GCash payment receipt upload step before concluding the execution sequence.

3.9 Data Dictionary

The Resort Administrator process flowchart represents the high-level governance and administrative oversight capabilities built into the web application. Accessing the control hub via Role-Based Access Control (RBAC) authentication, administrators can grant or revoke user privileges, adjust global cottage pricing parameters, and manage operational rules. The workflow concludes with the automated aggregation of system-wide transaction and reservation logs, allowing administrators to generate comprehensive financial audit reports and occupancy analytics for executive decision-making.

Table 3.1 Users Table

Field Name	Data Type	Constraint	Description
user_id	INIT	Primary Key, NOT NULL	Auto-incrementing unique user identifier.
full_name	VARCHAR (150)	NOT NULL	Full legal name of the user.
email	VARCHAR (100)	Unique, NOT NULL	Valid email address used for login and notifications.
password_hash	VARCHAR (255)	NULL	Bcrypt encrypted password hash.
contact_number	VARCHAR (20)	NULL	Optional mobile/phone contact details.
role	ENUM	NOT NULL	Access role: 'Guest', 'Staff', 'Admin'.
created_at	TIMESTAMP	NOT NULL	Record creation timestamp (Default: CURRENT_TIMESTAMP).

Table 3.2 Room_Amenity_Inventory Table

Field Name	Data Type	Constraint	Description
item_id	INIT	Primary Key, NOT NULL	Auto-incrementing unique item identifier.
item_name	VARCHAR (100)	NOT NULL	Accommodation or amenity name (e.g., Villa A, Cottage 1).
category	ENUM	NOT NULL	Item type: 'Villa', 'Cottage', 'Amenity'.
rate_per_day	DECIMAL (10,2)	NOT NULL	Standard daily rental rate or entry fee.
housekeeping_status	ENUM	NOT NULL	Unit status: 'Clean', 'Occupied', 'Maintenance'.
available_stock	INT	NOT NULL	Quantity available for booking/rent (Default: 1).

Table 3.3 Reservations Table

Field Name	Data Type	Constraint	Description
reservation_id	INIT	Primary Key, NOT NULL	Auto-incrementing unique reservation identifier.
user_id	INT	Foreign Key, NOT NULL	References users(user_id).
item_id	INT	Foreign Key, NOT NULL	References room_amenity_inventory(item_id).
check_in_date	DATE	NOT NULL	Scheduled check-in stay date.
check_out_date	DATE	NOT NULL	Scheduled check-out stay date.
booking_status	ENUM	NOT NULL	Workflow status: 'Pending', 'Confirmed', 'Rejected', 'Cancelled'.
total_amount	DECIMAL (10,2)	NOT NULL	Calculated total booking cost.

Table 3.4 Payment_Proofs Table

Field Name	Data Type	Constraint	Description
payment_id	INIT	Primary Key, NOT NULL	Auto-incrementing unique payment proof record identifier.
reservation_id	INT	Foreign Key, NOT NULL	References reservations(reservation_id).
receipt_image_path	VARCHAR(255)	NOT NULL	Relative file path to the uploaded digital receipt image.
reference_number	VARCHAR(100)	NOT NULL	Transaction reference number provided by guest/bank.
verification_status	ENUM	NOT NULL	Audit status: 'Pending', 'Approved', 'Rejected'.
uploaded_at	TIMESTAMP	NOT NULL	Receipt upload timestamp (Default: CURRENT_TIMESTAMP).

Table 3.4 Chat_Histories Table

Field Name	Data Type	Constraint	Description
chat_id	INIT	Primary Key, NOT NULL	Auto-incrementing unique chat log identifier.
user_id	INT	Foreign Key, NOT NULL	References users(user_id).
sender_type	ENUM	NOT NULL	Message source: 'Guest', 'Bot', 'Staff'.
message_text	TEXT	NOT NULL	Contents of the conversation message.
sent_at	TIMESTAMP	NOT NULL	Message dispatch timestamp (Default: CURRENT_TIMESTAMP).

3.10 System Development

The system development phase focuses on the technical realization of the Da Bali Resort Web System, transforming architectural models, entity-relationship schemas, and user interface designs into a fully functional web application. Development involves writing clean, modular source code across front-end user interfaces, server-side application logic, and relational database management systems. Throughout this phase, development tools, code repositories, and local environment stacks are utilized to construct, integrate, and verify core application features—such as user authentication, booking engine workflows, POS transactions, and administrative audit interfaces—prior to system deployment.

To ensure stability and maintainability, development is structured around a clear separation of concerns between client-side user experience and server-side business rules. The front-end interfaces are engineered to provide responsive layouts across both mobile devices and desktop terminals, ensuring seamless navigation for guests and staff. On the backend, server modules enforce crucial operational constraints, such as real-time date-overlap checking, receipt attachment validation, and automated PDF invoice generation. Database tables are systematically migrated, indexed, and linked with primary and foreign key constraints to uphold referential integrity across all transactional records.

Furthermore, continuous integration and testing practices are embedded directly within the development workflow. Source code is tracked using version control tools to manage feature updates, resolve bugs, and maintain historical revisions. Local development web servers and database instances are leveraged to execute functional

tests, API endpoint validations, and cross-browser compatibility checks. This structured development approach ensures that the resulting software is secure, scalable, and fully aligned with the operational requirements of Da Bali Resort.

3.11 Software Specification

The software specification outlines the specific application software, programming frameworks, database engines, and development tools required to build, run, and maintain the Da Bali Resort Web System.

Software Category	Tool / Technology	Purpose & Function
Operating System	Windows 11 / Linux (Ubuntu Server)	Host operating systems for development environments and production deployment.
Database Management System	MySQL / MariaDB	Relational database management system (RDBMS) executing table schemas, transaction logs, and queries.
Back-End Environment	Node.js / Express.js (or PHP 8.x)	Server-side runtime handling API routes, date-overlap validation logic, and session security.
Front-End Technologies	HTML5, CSS3, JavaScript (ES6+), React.js	Client-side technologies used to construct responsive web pages and interactive dashboards.
Security & Authentication	Bcrypt / JSON Web Tokens (JWT)	Hashing libraries and token protocols enforcing password encryption and Role-Based Access Control (RBAC).
Document	PDFKit / Dompdf	Server-side document

Processing		engine utilized for compiling and exporting printable PDF invoices and receipts.
Development & Testing Tools	Visual Studio Code, Postman, Git / GitHub	Code editor, API endpoint testing utility, and version control repository for managing system source code.

3.12 Software Specification

The hardware specification defines the physical computing equipment, memory capabilities, storage capacities, and processing power required to support development, administrative front-desk operations, and client access.

Server & Administrative Terminal Specifications

- **Processor (CPU):** Intel Core i5 / AMD Ryzen 5 (2.5 GHz Quad-Core or higher)
- **System Memory (RAM):** 8 GB DDR4 (16 GB Recommended)
- **Storage Capacity:** 256 GB Solid State Drive (SSD)
- **Network Device:** Gigabit Ethernet Adapter or Wi-Fi Module (Minimum 10 Mbps stable connection)
- **Display / Peripherals:** 19-inch Full HD Monitor (1920x1080 resolution), Keyboard, Mouse, and Document/Receipt Scanner.

Client Device Specifications (Guests)

- **Processor (CPU):** Dual-Core 1.8 GHz or equivalent ARM

mobile processor

- **System Memory (RAM):** Minimum 2 GB RAM (Mobile/Tablet) or 4 GB RAM (Desktop/Laptop)
- **Network Connectivity:** Active Mobile Data (4G/5G) or Wi-Fi internet connection
- **Display Interface:** Responsive screen display with touchscreen or standard pointer navigation.

3.14 Program Specification

The Program Specification outlines the functional features, operational constraints, technical scalability parameters, and cross-platform compatibility standards governing the Da Bali Resort Web System. These specifications define how the application operates, performs, adapts to growth, and interacts across different software environments.

3.13.1 Functional Requirements

Functional requirements define the specific actions, workflows, and operations that the web application must execute to support resort activities across different user roles.

- **User Authentication & RBAC Engine:** The system must provide secure account registration and login interfaces enforcing Role-Based Access Control (RBAC) across three distinct privilege levels (Guest, Staff, and Admin) with Bcrypt password hashing.
- **Reservation Engine & Date-Overlap Validation:** The system must process accommodation and cottage

bookings by executing a server-side validation algorithm that checks for scheduling overlaps against existing confirmed and pending reservations before allowing booking submissions.

- **Payment Upload & Verification Module:** The system must allow guests to upload digital payment proofs (e.g., GCash or bank receipts with reference numbers) for pending bookings, enabling staff and administrators to audit receipts, approve or reject transactions, and trigger status updates.
- **Point-of-Sale (POS) & Walk-In Management:** The system must provide front-desk staff with a dedicated interface to process immediate walk-in reservations, collect swimming pool entry fees, and record amenity rentals with real-time stock deductions.
- **Automated PDF Invoice Generation:** Upon confirming a reservation or POS transaction, the system must automatically generate downloadable and printable PDF receipts/invoices containing billing details and reference numbers.
- **Interactive FAQ Support Bot:** The platform must feature an automated chat interface capable of answering common guest inquiries regarding resort policies, pricing, and operating hours.

3.13.2 Non-Functional Requirements

Non-Functional requirements establish the quality, performance, usability, and security attributes of the system.

- **Security & Data Privacy:** All user passwords must be encrypted using Bcrypt hashing before database storage. Session handling must implement token-based or session-based security protocols to prevent unauthorized endpoint access and privilege escalation.
- **Performance & Response Time:** Server response times for database queries (such as date availability checks and login requests) must execute within less than two seconds under normal operational loads.
- **Usability & Interface Accessibility:** The system interface must adhere to responsive web design principles, featuring clear navigation menus, readable typography, intuitive button placements, and visual status indicators (e.g., pending, confirmed, clean, occupied).
- **Reliability & Data Integrity:** The relational database must maintain transactional integrity by enforcing primary key constraints, foreign key referential rules, and validation checks to prevent orphaned or duplicated records.

3.13.3 Scalability Specifications

Scalability specifications ensure that the application architecture can accommodate increasing volumes of operational data and concurrent web traffic over time.

- **Database Scalability:** The relational database schema is designed with indexed foreign keys, optimized query structures, and modular table separation to efficiently process historical booking logs, customer profiles, and transaction records without significant query

degradation.

- **Modular Code Architecture:** The software architecture adopts a decoupled, component-based layout separating front-end presentation, backend business logic routes, and database models. This structure allows new resort features—such as online food ordering or expanded cottage types—to be integrated without rewriting core code.
- **Resource Optimization:** Server-side scripts utilize asynchronous processing for database calls and background tasks, ensuring that CPU and memory utilization remain within optimal thresholds as concurrent user sessions increase during peak resort seasons.

3.13.4 Compatibility Requirements

Compatibility requirements define the cross-platform standards ensuring consistent application performance across various hardware client devices and software environments.

- **Cross-Browser Compatibility:** The web system must function consistently across all modern, standard web browsers, including Google Chrome, Mozilla Firefox, Microsoft Edge, and Apple Safari.
- **Cross-Platform Device Adaptability:** The application front-end must utilize fluid, responsive grid layouts that automatically adjust to desktop monitors (1080p and above), tablets, and mobile smartphones (iOS and Android).

- **Operating System & Server Environment:** The backend and database must be fully deployable across both Windows Server and Unix/Linux (e.g., Ubuntu LTS) operating environments using standard web hosting stacks.

3.14 Research Environment

The research study was conducted at Da Bali Resort, a premier leisure and hospitality establishment situated in Barangay Rosario, Municipality of Balingasag, Province of Misamis Oriental. Nestled within a growing commercial and eco-tourism corridor in Northern Mindanao, the resort offers guest accommodations, cottage rentals, swimming pool facilities, and event venues catering to local tourists, walk-in visitors, and private bookings. The geographical and operational setting of the resort provides an ideal environment for investigating the transition from traditional, manual booking processes to a modern, web-based digital management system.

The physical environment of Da Bali Resort consists of an administrative front-desk area, customer service counters, multiple cottage units, guest accommodations, and recreational facilities. The front desk serves as the primary operational hub where staff manage day-to-day guest accommodations, handle walk-in admissions, process manual booking records, and issue physical receipts. Conducting the study within this physical environment allowed the researchers to observe firsthand the operational challenges associated with paper-based scheduling, peak-season front-desk congestion, manual inventory tracking, and payment verification workflows.

From a technological perspective, the research environment encompasses both on-site administrative hardware and the broader

digital infrastructure accessible to prospective guests. The resort's front-desk operational environment is equipped with computer hardware terminals, Internet connectivity, and digital payment acceptance tools (such as GCash QR displays) used to record daily guest transactions. Concurrently, the virtual research environment extends to online users who access the resort's digital interfaces using mobile smartphones, tablets, and desktop computers over public wireless and cellular networks.

Conducting the research within Da Bali Resort provided a controlled yet authentic testing ground for evaluating the web-based system in real-world scenarios. It enabled direct interaction with key stakeholders—including resort administrators, front-desk staff, and visiting guests—during the data-gathering, system deployment, and user acceptance testing phases. This setting ensured that the developed software directly addresses localized operational requirements, enhances guest accessibility, and streamlines resort management efficiency.

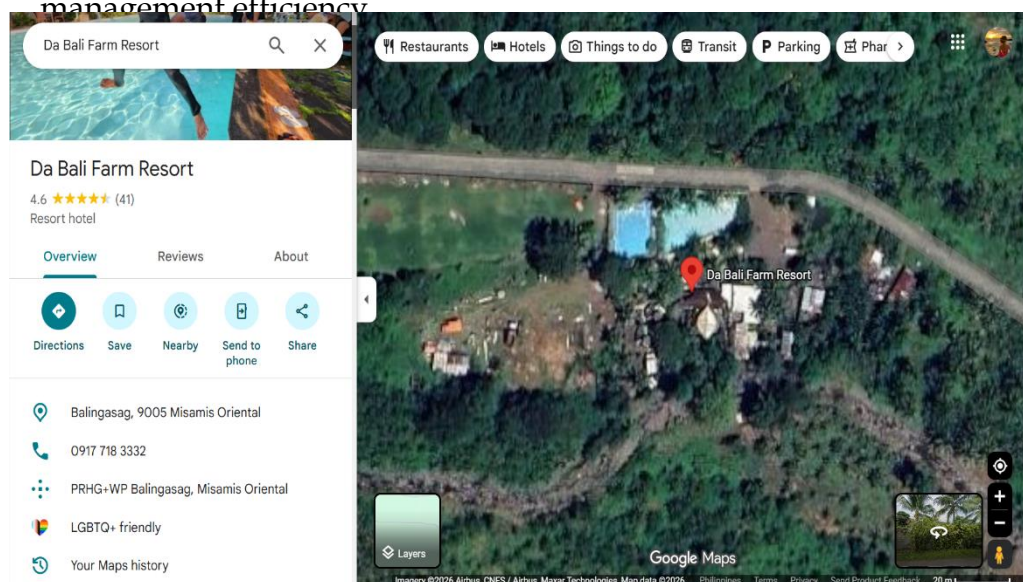


Figure 3.9 Location of Da Bali Resort in Barangay Rosario, Balingasag, Misamis Oriental

3.17 Unit of Analysis, Respondents, and Sampling

3.15.1 Unit of Analysis

The unit of analysis for this study is the Da Bali Resort Web System, specifically focusing on its functional performance, operational usability, processing security, and overall effectiveness in managing resort operations. The evaluation analyzes the system's core capabilities, including automated reservation workflows, date-overlap validation, real-time room and amenity inventory updates, digital payment verification, front-desk point-of-sale transactions, and administrative audit features.

3.15.2 Respondents of the Study

The target respondents of the study consist of a total of 10 selected evaluators directly involved in the operation, management, or end-user evaluation of the Da Bali Resort Web System. These respondents are divided into three distinct categories: one (1) Resort Administrator/Manager responsible for high-level system oversight and financial reports, two (2) Front-Desk Staff members who handle daily reservation confirmations, amenity inventory, and POS operations, and seven (7) Resort Guests who interact with the web platform to browse accommodations, submit online booking requests, and upload payment receipts.

3.15.3 Sampling Technique

To select the appropriate respondents for evaluating the system, a Purposive Sampling technique was employed. This non-probability sampling method was chosen to purposefully select individuals who possess direct experience with resort operations or active engagement with the web application interface. A sample size consisting of resort administrators,

front-desk personnel, and visiting guests was selected to ensure comprehensive feedback across technical functionality, administrative workflow integration, and end-user customer experience during user acceptance testing.

3.17 Data Source, Data Gathering Instruments, and Procedure

3.16.1 Data Source

The study utilizes both primary and secondary data sources to support the requirements analysis, system architecture design, database construction, and final software evaluation of the Da Bali Resort Web System. Primary data sources serve as the core foundation for evaluating system performance, usability, and operational alignment. This primary information was gathered directly from the ten (10) targeted respondents – comprising one (1) resort administrator, two (2) front-desk staff members, and seven (7) resort guests – through structured user acceptance testing, functional software evaluations, and preliminary interviews regarding resort operations.

Secondary data sources were gathered to establish domain knowledge, inform system workflows, and align technical features with established resort policies. These secondary materials include existing physical record-keeping documents from Da Bali Resort, such as paper booking logs, manual guest reservation books, physical receipt receipts, and published accommodation pricing catalogs. Additionally, academic literature, software engineering documentation, ISO quality standards, and database architecture references were reviewed to guide backend system construction and statistical analysis.

By combining primary end-user feedback with secondary operational documentation, the development team ensured that the web-based system directly resolves physical bottlenecks while adhering to rigorous software engineering standards. The primary evaluation responses provided empirical metrics regarding software quality, while secondary data ensured that all database schemas, user privilege levels, and administrative audit features accurately mirror the daily operational realities of the resort.

3.16.2 Data Gathering Instruments

To systematically collect qualitative and quantitative data throughout the system development life cycle, specific research instruments were constructed and utilized by the researchers. The primary instrument employed for software evaluation was an adapted evaluation questionnaire based on the ISO/IEC 25010 Software Quality Model. This structured instrument measures software performance across key quality characteristics—including Functional Suitability, Performance Efficiency, Usability, Reliability, Security, Maintainability, Portability, and Compatibility—using a 5-point Likert scale ranging from Strongly Agree to Strongly Disagree.

To complement the quantitative software evaluation, a structured interview guide was utilized during the initial requirements analysis phase. This instrument consisted of open-ended questions administered to resort management and front-desk personnel to document existing operational challenges, identify flaws in paper-based booking logs, determine user role privileges, and map out payment verification workflows. The insights gathered from this instrument directly informed the functional requirements,

entity-relationship schemas, and front-desk POS interface designs.

Finally, a technical system testing checklist was designed to evaluate backend application logic and database integrity during the development phase. This instrument enabled the researchers to execute test cases covering user authentication, Bcrypt password encryption, real-time date-overlap checking algorithms, file attachment validation for payment proofs, and automated PDF invoice generation. Together, these research instruments provided a comprehensive framework for validating both administrative utility and technical system performance.

3.16.3 Data Gathering Procedure

The data gathering procedure outlines the systematic, step-by-step process executed by the researchers to collect operational data, develop the web system, and evaluate its software quality. To ensure data accuracy and operational alignment with Da Bali Resort, the procedure spans from initial administrative clearance and requirements analysis to technical development, user acceptance testing, and statistical evaluation.

Step 1: Request for Permission

The research process initiated with a formal request for permission submitted by the researchers to the management of Da Bali Resort in Barangay Rosario, Balingasag, Misamis Oriental. Securing official administrative approval ensured authorized access to resort facilities, operational records, front-desk personnel, and visiting guests necessary for conducting interviews, operational observations, and system testing.

Step 2: Requirements Gathering

Following administrative approval, the researchers conducted structured interviews and direct observational studies with resort management and front-desk staff. During this phase, existing paper booking logs, room inventory schedules, physical receipt records, and daily front-desk workflows were examined to identify operational bottlenecks and define functional system requirements, user role access levels, and database entities.

Step 3: System Design and Development

Using the gathered operational data, the researchers designed and constructed the web application. Architectural models, relational database schemas, server-side API endpoints, and responsive client-side interfaces were developed to build core application modules, including real-time date-overlap validation, digital payment proof uploads, front-desk POS features, and automated PDF invoice generation.

Step 4: System Testing

Prior to end-user evaluation, the software underwent rigorous system testing using a technical checklist. The development team executed unit, integration, and security tests to verify backend route logic, Bcrypt password encryption, database foreign key constraints, receipt file handling, and cross-browser responsiveness across desktop and mobile devices.

Step 5: System Evaluation

After achieving functional stability, the web system was deployed in a staging environment for User Acceptance Testing (UAT). The application was presented to the ten (10) targeted respondents – comprising one (1) resort administrator, two (2)

front-desk staff members, and seven (7) resort guests—allowing them to interact with the platform and perform sample booking, auditing, and administrative tasks.

Step 6: Data Collection

Immediately following the system demonstration and hands-on testing session, the researchers administered the structured software evaluation questionnaire based on the ISO/IEC 25010 Software Quality Model to the respondents. The survey collected quantitative ratings across key quality attributes, alongside qualitative feedback regarding user experience and system functionality.

Step 7: Data Analysis

In the final step, all completed questionnaires were gathered, tabulated, and processed using statistical methods. Weighted mean values were calculated for each ISO/IEC 25010 software quality metric and interpreted using a defined Likert scale table to determine the overall effectiveness, usability, and operational suitability of the Da Bali Resort Web System.

3.17 Mode of Analysis and Statistical Instrument Procedure

3.17.1 Mode of Analysis

The data collected from the software evaluation questionnaires were compiled, organized, and subjected to descriptive statistical analysis. Descriptive statistics were utilized to summarize, analyze, and interpret the quantitative ratings provided by the ten (10) evaluators in accordance with the ISO/IEC 25010 Software Quality Model. The analysis assesses system performance across the standard's core quality characteristics: Functional Suitability, Performance Efficiency,

Usability, Reliability, Security, Maintainability, Portability, and Compatibility.

To determine the average evaluation score for each indicator and ISO/IEC 25010 quality dimension, the Weighted Mean ($\bar{X}$) was calculated using the following statistical formula:

$$\bar{X} = \frac{\sum(f \cdot x)}{N}$$

Where:

- $\bar{X}$ = Weighted Mean
- f = Frequency of responses for each scale rating
- x = Weight assigned to each scale response (1 to 5)
- $\sum(f \cdot x)$ = Sum of the products of frequency and weight
- N = Total number of respondents ($N = 10$)

Likert Scale Interpretation

The calculated weighted means for each ISO/IEC 25010 software quality parameter were interpreted using a standard 5-point Likert Scale Continuum. The numerical ranges, descriptive ratings, and operational interpretations are detailed in Table 3.1 below:

Scale Range	Rating	Descriptive Interpretation	Operational Meaning
4.21 – 5.00	5	Strongly Agree	The system fully meets and exceeds ISO/IEC 25010 software quality standards and operational requirements.

3.41 – 4.20	4	Agree	The system satisfies ISO/IEC 25010 quality standards with minor acceptable variations.
2.61 – 3.40	3	Neutral	The system meets basic quality requirements but requires moderate software or operational improvements.
1.81 – 2.60	2	Disagree	The system fails to meet key ISO/IEC 25010 operational quality requirements and needs major revisions.
1.00 – 1.80	1	Strongly Disagree	The system fails completely across functional, technical, or usability quality criteria.

3.17.2 Statistical Instrument Procedure

The statistical instrument procedure details the systematic workflow executed during User Acceptance Testing (UAT) to administer the evaluation forms, collect respondent scores ethically, tabulate data, and compute statistical measures for software evaluation.

Step 1: Administration of the Evaluation Instruments

The process began with the researchers briefing the ten (10) targeted evaluators (1 Resort Administrator, 2 Front-Desk Staff, and 7 Resort Guests) regarding the objectives of the evaluation. Following a live demonstration and hands-on interaction with the Da Bali Resort

Web System, the researchers distributed the structured evaluation forms based on the ISO/IEC 25010 Software Quality Model to gather feedback on system functionality, usability, performance, and security.

Step 2: Ethical Considerations

Prior to collecting feedback, strict ethical standards were implemented to protect respondent privacy and data confidentiality. Participants were informed that their participation was entirely voluntary and that they held the right to withdraw at any point. No personally identifiable information was gathered; all responses were anonymized, stored securely, and utilized strictly for academic research and system improvement purposes.

Step 3: Completion of the Questionnaires

The respondents evaluated the web platform by completing the ISO/IEC 25010 evaluation questionnaire. Participants scored each statement based on their direct experience navigating the resort system using a 5-point Likert scale, where ratings ranged from 5 (Strongly Agree) to 1 (Strongly Disagree). The researchers remained present to address any technical clarifications regarding questionnaire items without influencing respondent ratings.

Step 4: Tabulation of Responses

Once completed, all questionnaire forms were systematically gathered, audited for completeness, and coded into numerical data. The responses were organized into a centralized statistical matrix categorized by user groups (Administrator, Staff, and Guests) and sorted under each corresponding ISO/IEC 25010 software quality parameter.

Step 5: Statistical Analysis

The tabulated numerical data were processed using descriptive statistical formulas. Individual weighted means were calculated for each specific metric, followed by overall weighted means for each ISO/IEC 25010 characteristic. Finally, a composite Grand Mean was derived across all quality parameters to quantify the overall level of technical performance and user acceptance.

Step 6: Interpretation of Results

In the final step, the computed weighted mean values and overall Grand Mean were cross-referenced against the defined Likert Scale Continuum table. The numerical ranges were mapped to their corresponding descriptive ratings (e.g., Strongly Agree, Agree) and operational meanings, providing an empirical basis for assessing the software's readiness, quality, and operational suitability for Da Bali Resort.

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